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Quality 4.0 in Smart City Management

Frameworks, Applications and
Global Lessons

RADOSŁAW WOLNIAK
AND JOANNA ROSAK-SZYROCKA



Quality 4.0 in Smart City Management

This work introduces Quality 4.0 as an era of advanced quality management principles applied in the context of smart cities, blending classical and new approaches incorporating digital technologies, data analytics, artificial intelligence, and participatory system of governance. The authors define Quality 4.0 in the smart city as a complex socio-technical system where high-quality public services emerge from the interplay of data, institutions, and infrastructure in relation to citizens and accountability mechanisms.

This book proposes a new interpretative framework for Quality 4.0, targeting its usefulness to operationalizing urban strategies, enabling decision-making and adaptive systems of public management. The focus areas include system interoperability, security and data quality, public confidence and trust in the delivery systems, digital inclusion, urban resilience, and institutional adaptation and continuous learning. Its main contribution is in the framing of citizens as co-creators of quality public service. In the smart city ecosystem, urban user experience, social participation, digital justice, and transparency of decision-making are placed in the middle of the quality management process. From this perspective, Quality 4.0 is an instrument of correlation between operational efficiency and social responsibility, sustainable development, and democratic legitimacy.

The work represents an important contribution to the field of quality management, digital transformation in public administration, and smart city development. Due to its interdisciplinary nature, it will find a wide audience among researchers, public decision-makers, urban practitioners, and quality managers, as well as all those interested in the future of sustainable and responsible urban governance.

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and Joanna Rosak-Szyrocka**

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Introduction

Today modern smart city is complex socio-technical system, where technology, data, infrastructure, public institutions, and citizens coexist together to ensure the growing impact of these digital transformation initiatives on quality of urban services. It includes such solutions as the Internet of Things (IoT), artificial intelligence (AI), big data, digital twins, management dashboards, and predictive analytics that enhance monitoring, assessment, and optimization of urban processes. Using standards, accountability, data security, resilience, and social participation to ensure a coherent quality management system particularly demonstrates the value of these capabilities.

This monograph presents Quality 4.0 in Smart City Management as an integrated approach to the quality management process in smart cities. Quality 4.0 is perceived as a modern decision-making paradigm that integrates traditional quality tools with advanced digital technologies and the continuous learning logic of the urban system. This methodology lays the grounds for a process-oriented and systemic rationale of quality based on ISO standards, Total Quality Management, Six Sigma, and Plan–Do–Check–Act (PDCA) cycles while reinforcing the city altogether with IoT, AI, big data, digital twins, and dashboards to be monitored adaptively in real time with predictive analytics.

The discussion begins with the premise that developing smart cities requires a holistic quality management model that links urban strategy and operational processes, metrics, data security and resilience, and citizen experience development. Cities create more data and build digital public services and smart infrastructure; hence, they need systems to integrate multiple complex areas as well as mechanisms for continued institutional coordination, privacy protection, trust-building, and public value creation. The areas outlined in connection with it are organized by Quality 4.0, enabling quality to be perceived as a multilevel and dynamic category involved across the urban management processes. The monograph consists of seven chapters. Chapter 1 proposes Quality 4.0 as a smart city management framework, where traditional quality standards fit into a digital multi-stakeholder dynamic urban environment. Chapter 2 outlines the major motivating conditions for Quality 4.0

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like data privacy, cybersecurity, financial resources, interoperability, and technological complexity. Chapter 3 emphasizes citizens, trust, participation, inclusiveness, and digital equity as essential aspects of the quality of public services. Chapter 4 analyzes the articulation between KPIs (Key Performance Indicators), models of evaluation, data quality, and digital security. Chapter 5 describes practical cases of various smart city governance models around the world from Eastern Europe and the Global South. Future Trends such as AI, digital twins, edge computing, autonomous systems, resilience, and sustainability are covered in Chapter 6. In Chapter 7 authors sum up key conclusions and provide recommendations on the future development of Quality 4.0 as a framework for management of smart cities. The monograph recommends an innovative Smart City Quality 4.0 Continuous Learning Model presented in Figure 0.1.

Quality 4.0 in smart city management is defined in the model as a consequence of data, quality-oriented ecosystems, cybersecurity and privacy, resilience and sustainability, interoperability, and citizen experience. And at the same time, it suggests urban quality as continuous system learning behavior that requires not only constant monitoring and analysis but also feedback loops, adaptation, and therefore also improvement of public services.

The proposed approach conceives a smart city as one that scores highly on quality through its capacity to reconcile the technological efficiency of institutions with institutional responsibility concerning social value.

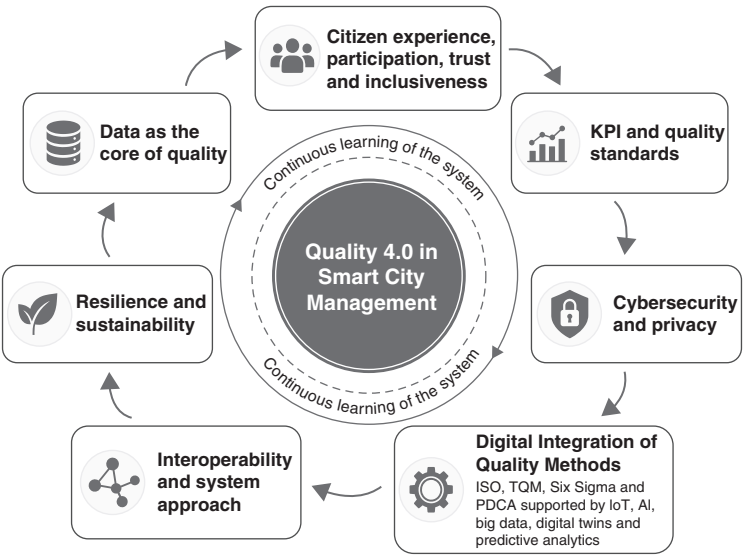


Figure 0.1 Smart City Quality 4.0 Continuous Learning Model.

The components of smart city quality include process efficiency, security and privacy, transparency in decision-making, service accessibility and affordability, infrastructure resilience, environmental sustainability, digital equity, and citizen trust. Quality, understood in this latter sense, is not just a matter of measuring individual outcomes but rather the ability of the city to learn and respond to changes in social, technological, or environmental conditions and adjust its services accordingly.

The importance of gathering lessons from cities is also emphasized by the monograph. The analyzed cases show that Quality 4.0 evolves in response to local institutional, technological, social, and financial factors. For this reason, particular importance shall be given to build up the data capacity of the city for the responsible use of data, strengthen inter-organizational coordination capacities, enhance citizen engagement, and systematically improve public services. Therefore, Quality 4.0 may be perceived as a remedy to the challenges of the real-time city where data, digital technologies, institutions, and citizen co-generate an ecosystem of urban governance.

As Calabrese, Kloeckl and Ratti (2008) noted, “The real-time city is now reality.” Hence, Quality 4.0 provides a scaffolding for shaping the real-time city into an urban system that learns, cares, secures, and responds to its citizens.

We hope that this book becomes a relevant resource where the reader can gather knowledge to build alternative understanding and progress in Quality 4.0 for managing smart cities.

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